CloudFlix – Scalable Netflix-Style Video Streaming Platform on AWS

🟣 2. Overview

CloudFlix is a cloud-native video streaming platform designed using **microservices architecture** and deployed on AWS.

The system enables users to upload, process, and stream videos efficiently using a **scalable, event-driven architecture** similar to modern OTT platforms.

🟣 3. Key Features

- User authentication using JWT
- Video upload and storage
- Automated video processing pipeline
- HLS adaptive streaming (.m3u8)
- Secure streaming using signed URLs
- Watch history tracking
- Scalable microservices backend
- CDN-based global content delivery

🟣 4. Architecture Overview

The platform is built using a **distributed system design** consisting of:

- API Gateway (central entry point)
- Microservices (Auth, Video, etc.)
- Event-driven processing pipeline
- CDN-based video delivery
- Multi-database architecture

🟣 5. System Architecture Flow

🎥 Video Upload Flow

User → API Gateway → Video Service → S3 Upload

S3 → Lambda → SNS → SQS → Worker → MediaConvert → S3 (HLS Output)

▶️ Video Streaming Flow

User → CloudFront → S3 (HLS files)

📊 Data Flow

- Metadata → RDS
- Watch History → DynamoDB
- Cached Data → Redis

🟣 6. AWS Services Used

☁️ Compute

- Amazon EC2
- AWS Lambda
- Auto Scaling Group
- Application Load Balancer

🗄️ Storage

- Amazon S3 (Video storage & HLS output)

🎥 Video Processing

- AWS MediaConvert

🌐 CDN

- Amazon CloudFront (with signed URLs)

🗃️ Databases

- Amazon RDS (Aurora – relational data)
- Amazon DynamoDB (NoSQL – watch history)
- Amazon ElastiCache (Redis – caching)

📩 Messaging

- Amazon SNS
- Amazon SQS

🌍 Networking

- Amazon VPC
- NAT Gateway
- Transit Gateway
- Public & Private Subnets

📊 Monitoring

- Amazon CloudWatch

🟣 7. Microservices Architecture

The system is divided into independent services:

1. Auth Service

- User registration & login
- JWT token generation

2. Video Service

- Upload videos
- Store metadata
- Manage video records

3. Streaming Service

- Provide secure video URLs
- Handle playback

4. Worker Service

- Process video jobs
- Trigger MediaConvert

🟣 8. Event-Driven Pipeline

The system uses asynchronous processing:

S3 Upload → Lambda Trigger → SNS → SQS → Worker → MediaConvert

Benefits:

- Loose coupling
- High scalability
- Fault tolerance

🟣 9. Database Design

RDS (Aurora)

Stores structured data:

- Users
- Videos
- Metadata

DynamoDB

Stores:

- Watch history
- User activity

Redis (ElastiCache)

Used for:

- Caching popular videos
- Fast data retrieval

🟣 10. Security

- JWT-based authentication
- IAM roles and policies
- Signed URLs for secure video streaming
- Private subnets for backend services
- Restricted S3 access

🟣 11. Networking Architecture

- Multi-VPC architecture
- Transit Gateway connects all VPCs
- Public VPC → Load Balancer
- Application VPC → Microservices
- Data VPC → Databases

🟣 12. Scalability & High Availability

- Auto Scaling for EC2 instances
- Load Balancer for traffic distribution
- CDN (CloudFront) for global delivery
- Queue-based processing (SQS)

🟣 13. Development Phases

Phase 1:

Local development with Node.js

Phase 2:

Integration with S3

Phase 3:

Deployment on EC2

Phase 4:

Advanced AWS architecture

- Lambda
- CloudFront
- MediaConvert
- Multi-VPC

🟣 14. Folder Structure

```
cloudflix/
│
├── backend/
│   ├── auth-service
│   ├── video-service
│   ├── worker-service
│
├── gateway/
│   └── api-gateway
│
├── frontend/
│   └── flutter-app
│
└── infrastructure/
    └── diagrams
```

🟣 15. API Endpoints (Example)

Auth

- POST /auth/register
- POST /auth/login

Videos

- POST /videos/upload
- GET /videos
- GET /videos/play/:id

🟣 16. Learning Outcomes

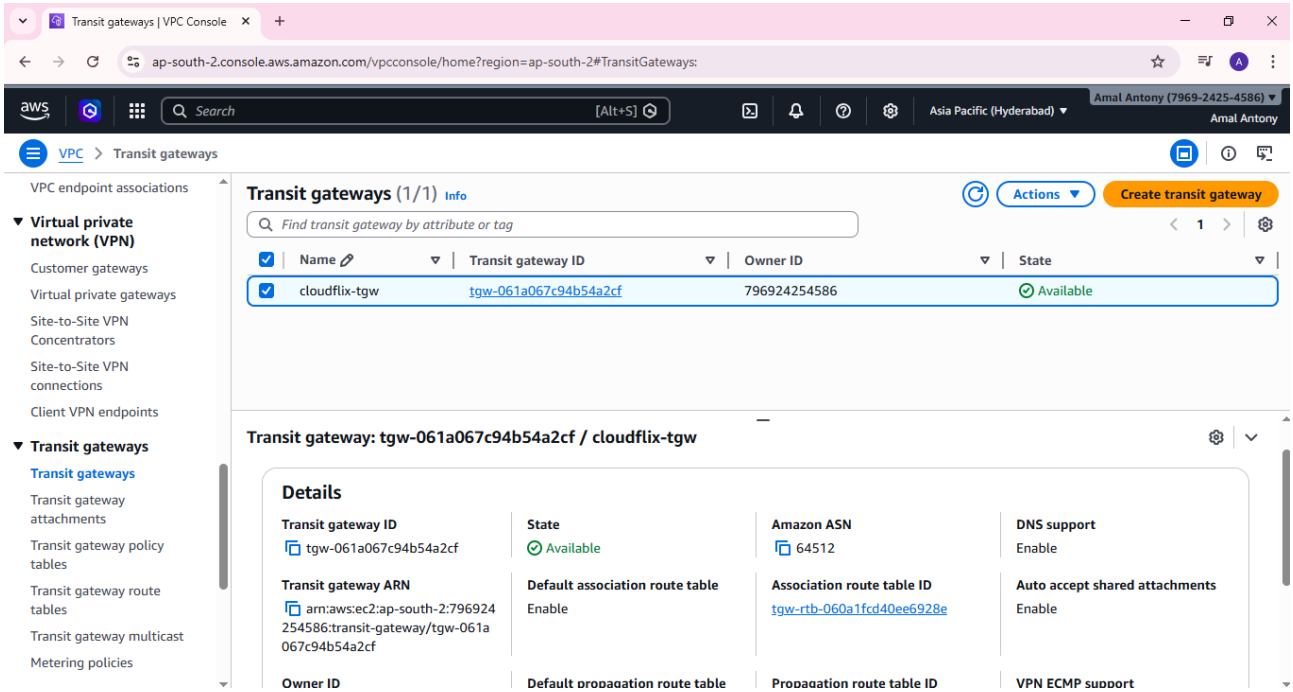
- Designed scalable cloud architecture
- Implemented event-driven systems
- Worked with AWS networking (VPC, NAT, TGW)
- Built production-level backend services
- Learned CDN-based streaming

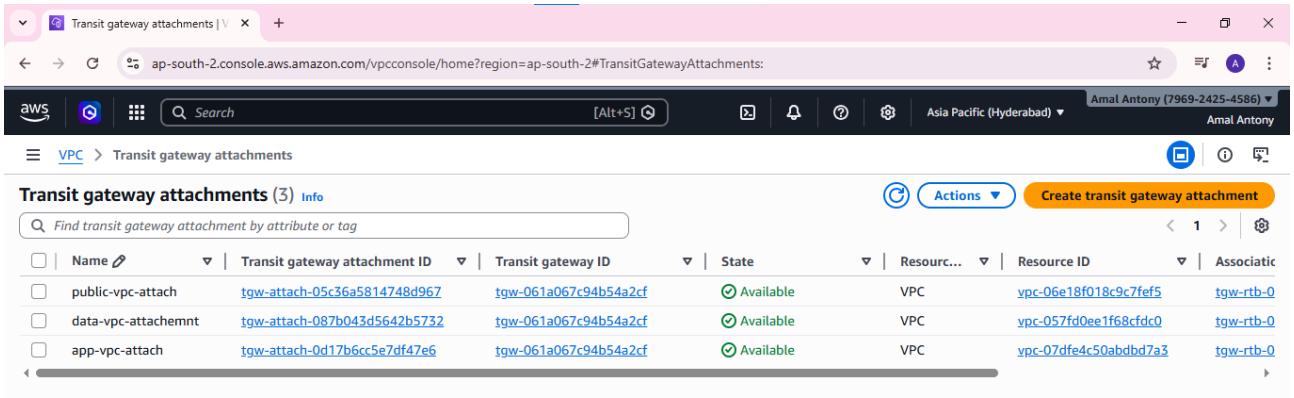
🟣 17. Conclusion

CloudFlix demonstrates how to build a **production-ready, scalable video streaming platform** using AWS.

It highlights real-world concepts such as:

- Microservices architecture
- Event-driven processing
- CDN-based delivery
- Distributed system design




Transit gateway attachments | V
ap-south-2.console.aws.amazon.com/vpcconsole/home?region=ap-south-2#TransitGatewayAttachments:
Search
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Asia Pacific (Hyderabad)
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VPC > Transit gateway attachments
Transit gateway attachments (3) Info
Actions
Create transit gateway attachment
Find transit gateway attachment by attribute or tag
Name
Transit gateway attachment ID
Transit gateway ID
State
Resourc...
Resource ID
Associatic
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tgw-attach-05c36a5814748d967
tgw-061a067c94b54a2cf
Available
VPC
vpc-06e18f018c9c7fef5
tgw-rtb-0
data-vpc-attachemnt
tgw-attach-087b043d5642b5732
tgw-061a067c94b54a2cf
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VPC
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tgw-rtb-0
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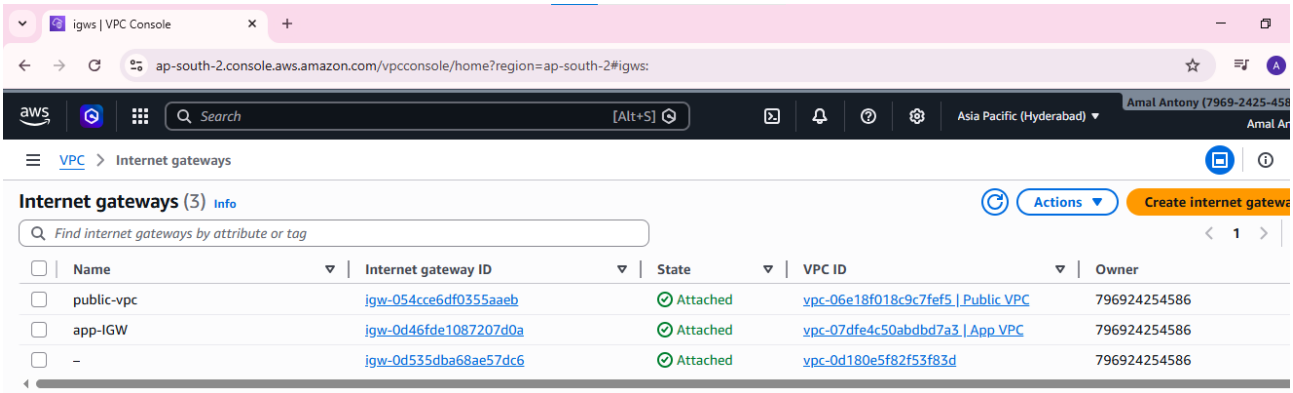
igws | VPC Console
ap-south-2.console.aws.amazon.com/vpcconsole/home?region=ap-south-2#igws:
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Asia Pacific (Hyderabad)
VPC > Internet gateways
Internet gateways (3) Info
Actions
Find internet gateways by attribute or tag
Name
Internet gateway ID
State
VPC ID
Owner
public-vpc
igw-054cce6df0355aaeb
Attached
vpc-06e18f018c9c7fef5 | Public VPC
796924254586
app-IGW
igw-0d46fde1087207d0a
Attached
vpc-07dfe4c50abdbd7a3 | App VPC
796924254586
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igw-0d535dba68ae57dc6
Attached
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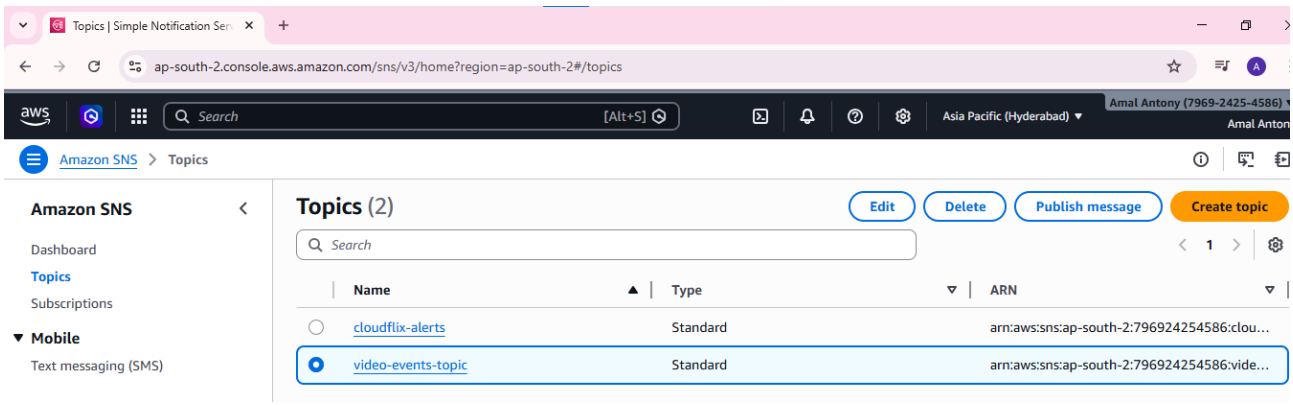
796924254586


Topics | Simple Notification Ser
ap-south-2.console.aws.amazon.com/sns/v3/home?region=ap-south-2#/topics
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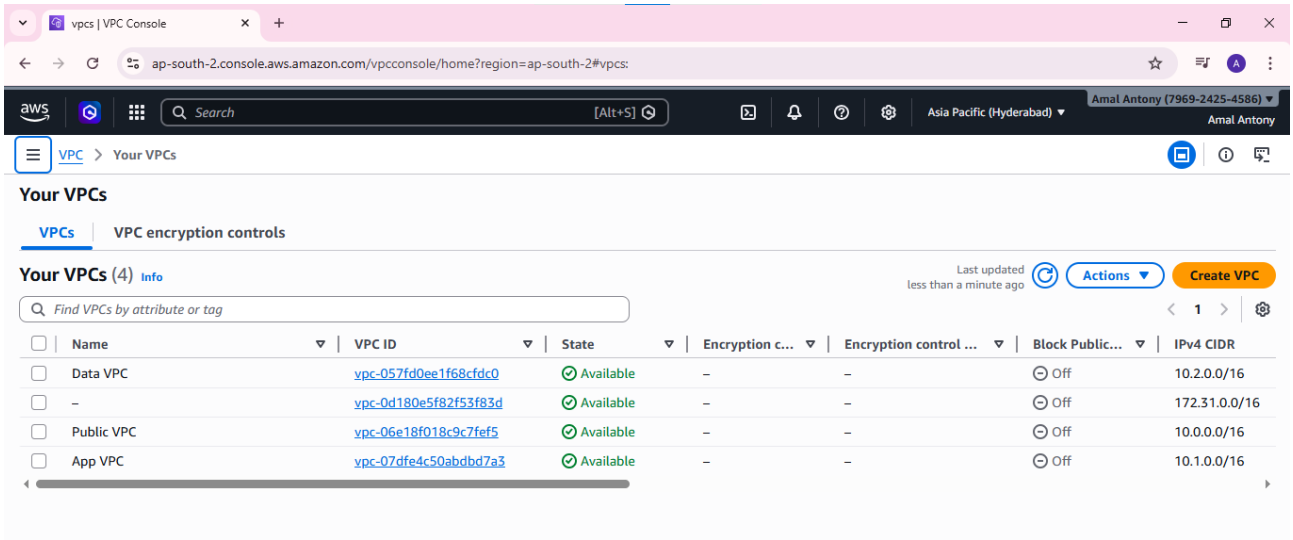
Amazon SNS > Topics
Amazon SNS
Dashboard
Topics
Subscriptions
Mobile
Text messaging (SMS)
Topics (2)
Edit
Delete
Publish message
Create topic
Search
Name
Type
ARN
cloudflix-alerts
Standard
arn:aws:sns:ap-south-2:796924254586:clou...
video-events-topic
Standard
arn:aws:sns:ap-south-2:796924254586:vide...


vpcs | VPC Console
ap-south-2.console.aws.amazon.com/vpcconsole/home?region=ap-south-2#vpcs:
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VPC > Your VPCs
Your VPCs
VPCs
VPC encryption controls
Your VPCs (4) Info
Last updated less than a minute ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Public...
IPv4 CIDR
Data VPC
vpc-057fd0ee1f68cfdc0
Available
Off
10.2.0.0/16
–
vpc-0d180e5f82f53f83d
Available
Off
172.31.0.0/16
Public VPC
vpc-06e18f018c9c7fef5
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App VPC
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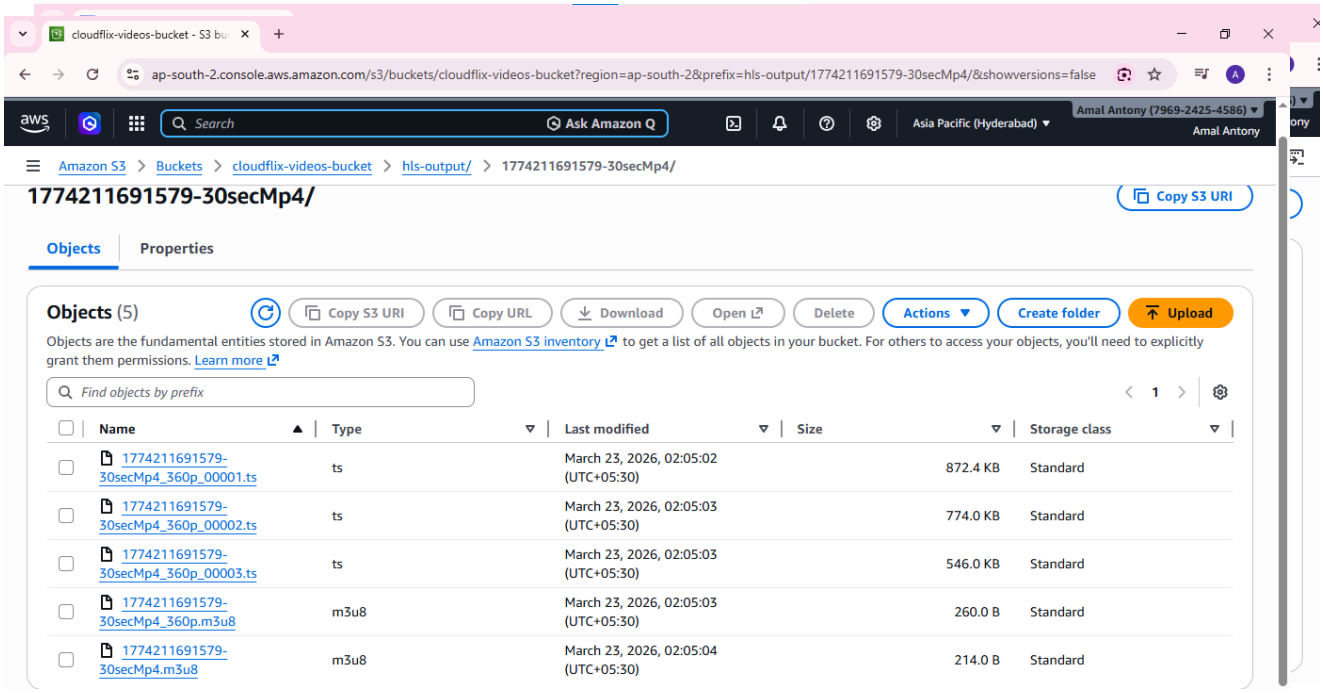
cloudflix-videos-bucket - S3 bu
ap-south-2.console.aws.amazon.com/s3/buckets/cloudflix-videos-bucket?region=ap-south-2&prefix=hls-output/1774211691579-30secMp4/&showversions=false
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Amazon S3 > Buckets > cloudflix-videos-bucket > hls-output/ > 1774211691579-30secMp4/
1774211691579-30secMp4/
Copy S3 URI
Objects
Properties
Objects (5)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
1774211691579-30secMp4_360p_00001.ts
ts
March 23, 2026, 02:05:02 (UTC+05:30)
872.4 KB
Standard
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ts
March 23, 2026, 02:05:03 (UTC+05:30)
774.0 KB
Standard
1774211691579-30secMp4_360p_00003.ts
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546.0 KB
Standard
1774211691579-30secMp4_360p.m3u8
m3u8
March 23, 2026, 02:05:03 (UTC+05:30)
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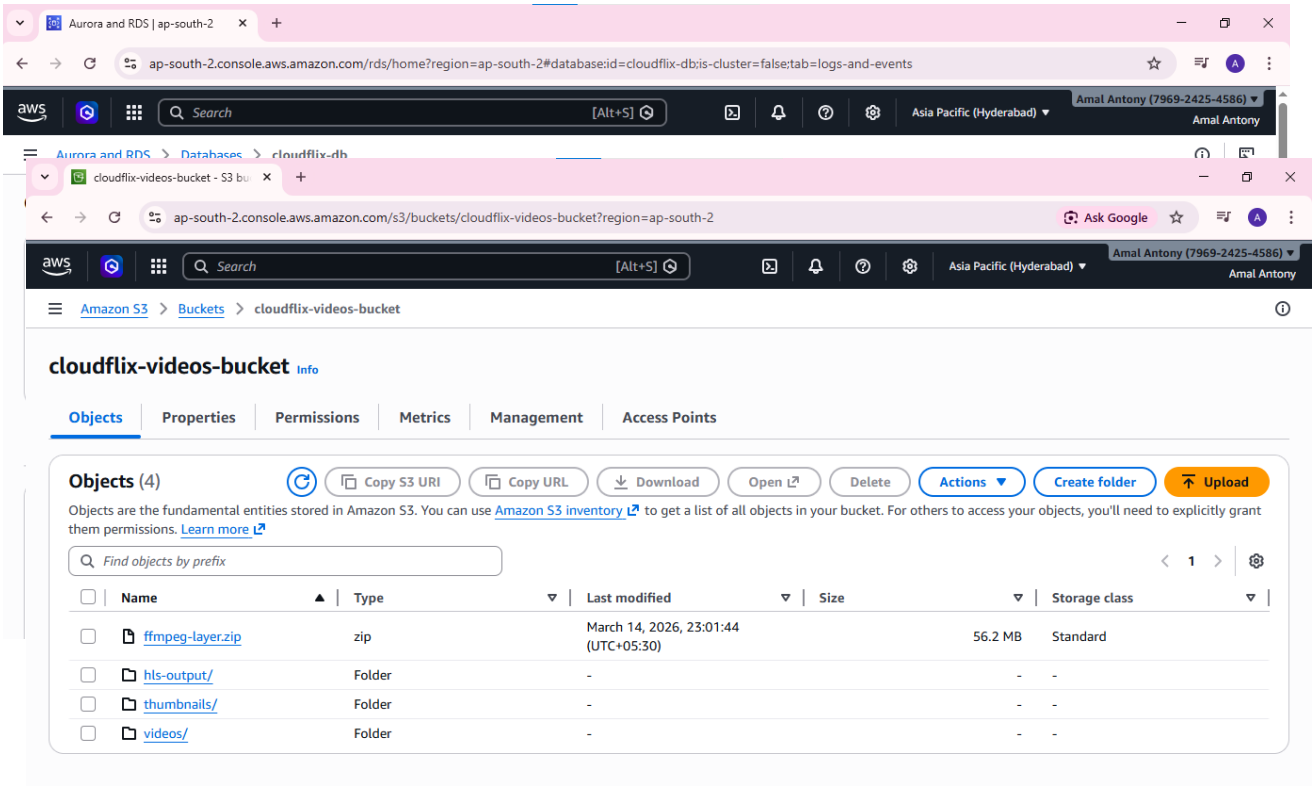
Standard


Aurora and RDS | ap-south-2
ap-south-2.console.aws.amazon.com/rds/home?region=ap-south-2#database:id=cloudflix-db;is-cluster=false;tab=logs-and-events
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Amal Antony
Amazon S3 > Buckets > cloudflix-videos-bucket
cloudflix-videos-bucket Info
Objects
Properties
Permissions
Metrics
Management
Access Points
Objects (4)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
1
Name
Type
Last modified
Size
Storage class
ffmpeg-layer.zip
zip
March 14, 2026, 23:01:44 (UTC+05:30)
56.2 MB
Standard
hls-output/
Folder
thumbnails/
Folder
videos/
Folder

Distributions | CloudFront | Glo

us-east-1.console.aws.amazon.com/cloudfront/v4/home?region=ap-south-1#/distributions/E3SHBLHPO0BQVN

Search [Alt+S] Global Amal Antony (7969-2425-4586)

CloudFront > Distributions > E3SHBLHPO0BQVN

s3cloudFront Free plan

View metrics

Details

Distribution domain name
d2nj2h7rya12fv.cloudfront.net

Billing
Free plan ($0/month)
Manage plan

ARN
arn:aws:cloudfront::796924254586:distribution/E3SHBLHPO0BQVN

Last modified
March 18, 2026 at 6:32:53 AM UTC

General | Security | Origins | Behaviors | Error pages | Invalidations | Logging | Tags

Settings

Edit

Name
s3cloudFront

Description
-

Price class
Use all edge locations (best performance)

Alternate domain names
-
Add domain

Standard logging
Available with the Pro plan

Default root object
-

View table | Amazon DynamoDB

ap-south-2.console.aws.amazon.com/dynamodbv2/home?region=ap-south-2#table?name=WatchHistory

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DynamoDB > Tables > WatchHistory

Tables (1)

Filter by tag: Any tag key
Filter by tag value: Any tag value
Find tables

1
WatchHistory

WatchHistory

Last updated March 23, 2026, 03:40 (UTC+5:30)

Actions | Explore table items

Settings | Indexes | Monitor | Global tables | Backups | Exports and streams | Permissions

General information Info

Get live item count

Partition key
userId (String)

Sort key
videoId (String)

Capacity mode
On-demand

Table status
Active

Alarms
No active alarms

Point-in-time recovery (PITR) Info
Off

Item count
2

Table size
126 bytes

Jobs | MediaConvert | ap-south

ap-south-1.console.aws.amazon.com/mediaconvert/home?region=ap-south-1#/jobs/list

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AWS Elemental MediaConvert > Jobs

AWS Elemental MediaConvert

- Jobs
- Output presets
- Job templates
- Queues
- Certificates

Jobs

Refresh | Cancel job | Import job | Create job

Status filter: Any status
Queue filter: Any queue

1

Job ID	First input file name	Submit time	Finish time	Status	Job percent complete	Queue
1774211696407-m3vn9z	1774211691579-30secMp4.mp4	2026-03-23 02:04:56	2026-03-23 02:05:04	COMPLETE		Default
1774211353375-rvrlh3	1774211349641-30secMp4.mp4	2026-03-23 01:59:13	2026-03-23 01:59:19	COMPLETE		Default
1773924367118-3455eo	1773924366225-30secMp4.mp4	2026-03-19 18:16:07	2026-03-19 18:16:12	COMPLETE		Default
1773924176873-yr6mem	1773924173649-30secMp4.mp4	2026-03-19 18:12:57	2026-03-19 18:13:02	COMPLETE		Default
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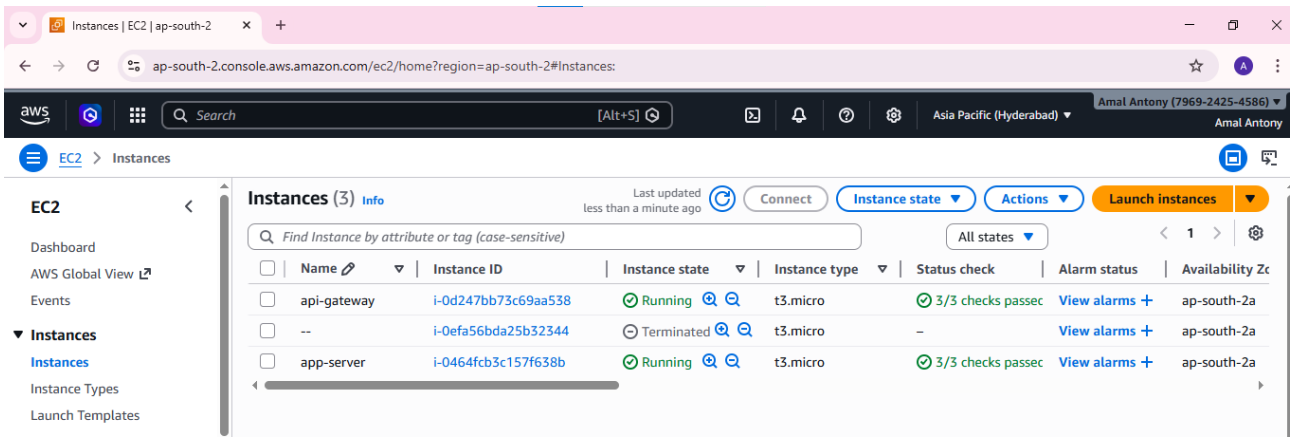
Instances | EC2 | ap-south-2
ap-south-2.console.aws.amazon.com/ec2/home?region=ap-south-2#Instances:
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EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Instances (3) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zo
api-gateway
i-0d247bb73c69aa538
Running
t3.micro
3/3 checks passed
View alarms
ap-south-2a
--
i-0efa56bda25b32344
Terminated
t3.micro
-
View alarms
ap-south-2a
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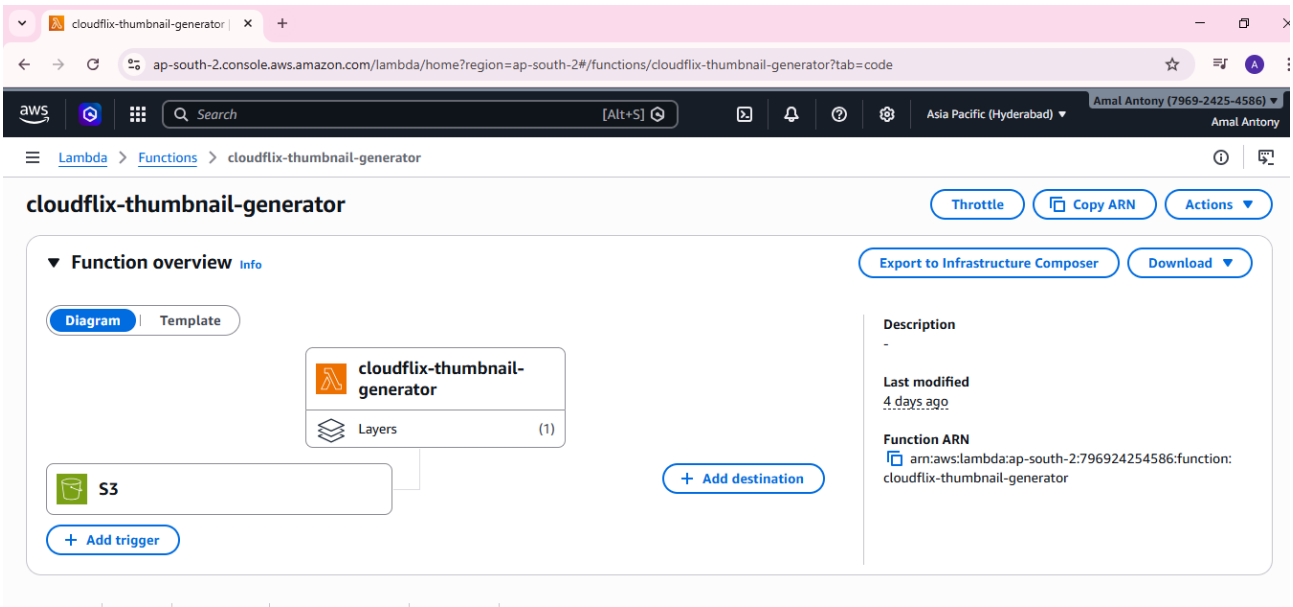
Running
t3.micro
3/3 checks passed
View alarms
ap-south-2a


cloudflix-thumbnail-generator
ap-south-2.console.aws.amazon.com/lambda/home?region=ap-south-2#/functions/cloudflix-thumbnail-generator?tab=code
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Lambda > Functions > cloudflix-thumbnail-generator
cloudflix-thumbnail-generator
Throttle
Copy ARN
Actions
Function overview Info
Export to Infrastructure Composer
Download
Diagram
Template
cloudflix-thumbnail-generator
Layers
(1)
S3
Add trigger
Add destination
Description
-
Last modified
4 days ago
Function ARN
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